S-T-540E
January 22, 1976
SUPERSEDING
Fed. Spec. S-T-540D
April 17, 1972

FEDERAL SPECIFICATION

TOASTERS, CONTINUOUS (CONVEYOR, HEAVY DUTY); ALL ELECTRIC AND COMBINATION GAS AND ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heavy duty, conveyor toasters for sliced bread and buns.

1.2 Classification. Toasters shall be of the following types and sizes, as specified (see 6.2).

Type I - Electric.

Type II - Gas.

Size 1 - Deleted (see 6.4).
Size 2 - 540 slices per hour capacity.
Size 3 - 720 slices per hour capacity (electric only).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-R-175	-	Resistance Wire.
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packaging and Packing.

FSC 7310

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Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-I-24092	-	Insulating Varnish, Electrical, Impregnating.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirement for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI)

Z21.31 - Standard for Gas Counter Appliances

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018 or the American Gas Association, Inc., 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

American Gas Association (A.G.A.)

A.G.A. Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

American Society for Testing and Materials (ASTM)

A 176 - Stainless Steel and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

B 407 - Nickel-Iron-Chromium Alloy Seamless Pipe and Tube

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Sanitation Foundation (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

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Underwriters' Laboratories, Inc. (UL)

UL 197 - Commercial Electric Cooking Appliances

UL 197 Supplement S-1 - Commerical Electric Cooking Appliance for Marine Use

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., NY 11746; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard product. The toasters delivered under this specification shall be the manufacturer's current commercial product, except for any changes necessary to comply with specification requirements. All toasters, of one type and size, furnished under this specification, on any one contract, including parts and assemblies thereof, shall be interchangeable.

3.2 Codes and standards. Toasters shall comply with the applicable requirements of ANSI Z21.31, NSF Standard No. 4, UL No. 197 and, when applicable for shipboard use, UL 197 Supplement S-1.

3.2.1 Compliance. Prior to commencing production, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the toaster he proposes to supply under this specification meet the applicable requirements of ANSI Z21.31, NSF Standard No. 4, UL 197 and, when applicable, UL 197 supplement S-1 as follows:

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3.2.1.1 ANSI. Acceptable evidence of meeting the applicable requirements of ANSI Z21.31 shall be a reproduced copy of the AGA Appliance Certificate, a listing in the latest A.G.A. "Directory of Certified Appliances and Accessories", or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating compliance with ANSI Z21.31.

3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished toaster, or

(2) A certification for the toaster issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the toasters are being procured, indicating that the toasters have been tested and conform to NSF Standard No. 4.

3.2.1.3 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label or UL listing mark or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the toaster offered conforms to the applicable requirements of UL 197 and when applicable, UL 197, Supplement S-1.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his standard commercial toasters, provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thicknesses.

3.3.1 Stainless steel sheet and strip. Stainless steel sheet or strip shall conform to type 430 steel specified in ASTM A 176.

3.3.2 Seamless metal tube. The seamless metal tube shall conform to ASTM B 407.

3.4 Design and construction. The type I toasters shall be electrically heated and the type II toasters shall be gas heated. The toasters shall have an electric motor driven conveyor system. The outer shell of the toaster shall have an opening for placement of bread or bun slices on conveyor racks. The toaster shall be provided with either a nonadjustable deflector or adjustable deflector located under the discharge end of the conveyor rack. When specified (see 6.2), a nonadjustable deflector shall be provided and shall deliver toast to the loading side of the toaster. The adjustable deflector shall deliver toasted items to either front or rear of the toaster. Adjustment of the adjustable deflector shall be accomplished without the use of tools, while the toaster is operating. A minimum of one removal pan shall be furnished with each toaster located to catch crumbs.

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3.4.1 Outer shell. The outer shell of the toaster shall be fabricated of stainless steel specified in 3.3.1, having a thickness of not less than 25 gauge (0.0218) inch. The outer shell shall inclose the electric motor, conveyor system, and heating chamber. The outer shell shall be provided with access panels or door(s) for repairs or replacement of parts.

3.4.2 Conveyor rack assembly. The conveyor rack assembly shall be driven by an electric motor and the electric motor shall be overload protected in the event the conveyor system jams. Conveyor racks shall carry the bread or bun slices between the heating elements in a manner to provide uniform toasting of 540 and 720 slices of bread per hour for sizes 2 and 3 toasters, respectively. An overrunning clutch or other means shall be provided in the drive mechanism to allow manual movement (see 4.4.4) of the conveyor racks. The relative position of support wires of the bread racks and conveyor chain studs shall be such that no interference shall occur when the chain driving mechanism and bread racks are positioned as close together as possible and the conveyor and bread racks rotated manually (see 4.4.3 and 4.4.4).

3.4.3 Electrical system.

3.4.3.1 Electric motors. Electric motors used on types I and II toasters shall conform to the following requirements:

Volts	See 3.5.1.1 and 3.5.2.1
Duty	Continuous
Enclosure	Drip proof
Ambient temperature	50°C.
Insulation	Class A, B, or H
Bearings	Ball or sleeve

Alternating current (ac) motors shall not be of the commutator type.

3.4.3.2 Terminal board or box. Unless otherwise specified (see 6.2), a terminal board or box shall be furnished within the toaster enclosure on type I toasters to provide for permanent connection to the installation wiring system. The terminal board or box shall have a ground terminal connected to the toaster outer shell.

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3.4.3.3 Cable and plug assembly. The type I toasters not supplied with a terminal board or box (see 3.4.3.2), and type II toasters shall be electrically grounded by a three wire grounding type cord and a three prong grounding attachment plug cap. The length of the cord for the type I toaster shall be not less than 3 feet nor more than 4 feet. The length of cord for type II toasters shall be not less than 3 feet or more than 6 feet.

3.4.3.4 Controls. An on-off switch shall be provided to control both the electric motor circuit and the electrical heating element circuit for the type I toasters and control the electric motor circuit for the type II toasters. The types I and II toasters shall be provided with means to obtain light, medium, or dark colored toast or bun halves (as desired) when tested in accordance with 4.4.2. Unless otherwise specified (see 6.2), for single side toasting a separate on-off switch or a set of contacts on the primary on-off switch to break the circuit to the rear heating element shall be provided. When switches are placed in the off position, all conductors of the power supply circuit lines shall be broken. When toggle switches are used guards shall be provided to prevent accidental operation.

3.4.3.5 Electric heating elements. Electric heating elements on type I toasters shall be nickel-chromium resistance wire or ribbon conforming to composition D or E of QQ-R-175, or the heating element shall be incased in a seamless metal tube as specified in 3.3.2, or any other type of manufacturer's commercial element. The heating elements shall be designed for only one method of installation so that they cannot be incorrectly positioned. Like elements shall be interchangeable.

3.4.3.6 Electromagnetic compatibility. When specified, (see 6.2), equipment procured under this specification shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.5).

3.4.3.7 Fungus and moisture resistance. When specified (see 6.2), components of the motor and other electrical equipment shall be treated to resist moisture and fungus, as follows:

Electrical components such as switches and fuses shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated. Other circuit elements having a temperature rise of not more than 40 C. operating at full load shall be coated with a fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, switchboards, terminal-end-junction blocks, and coils. Circuit elements, such as motor coils which have a temperature rise exceeding 40 C. when operating at full load, shall be given two coats of varnish conforming to applicable class of MIL-I-24092. The coats shall be applied to preclude entrapment of air or moisture.

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3.4.4 Gas fuel system. The fuel system of type II toasters, including burner(s), shall be designed to burn natural, and liquefied petroleum gas. Burner gas valves and pilot gas valves shall be of the fixed orifice type with interchangeable orifice hoods or bushings. Each burner shall be provided with two orifice hoods or bushings. One bushing or orifice hood drilled for natural gas and the other orifice hood or bushing drilled for liquefied gas. The manufacturer shall provide instructions for conversion from any one type of gas to the other. The orifice hoods or bushings shall be color coded as follows:

(1) Natural gas - unpainted brass
(2) Liquefied petroleum gas - black

Each type II toaster shall be provided with a filter in the gas line to the pilot. When specified (see 6.2) type II toasters shall be provided with an A.G.A. certified quick-disconnect assembly (socket and plug) and a heavy duty commercial-type flexible metal connector consisting of a male pipe thread fitting at one end and a swivel union, female pipe thread at the other end. The flexible metal connector shall have a minimum bend radius of not less than 2-1/2 inches. The quick-disconnect assembly shall be compatible with the flexible connector male thread fitting. The quick-disconnect assembly shall be located at the gas supply line. The swivel union shall be compatible with the inlet gas fitting on the unit. Means shall be provided to secure the flexible metal connector to the back of the unit when disconnected from the supply line. Unless otherwise specified (see 6.2), the length of gas-line connector shall be 60 inches.

3.4.4.1 Automatic ignition. Type II toasters main burners shall be equipped with means for automatic ignition of gas. A pilot, including a 100 percent shut-off device shall be provided for each toaster burner. The pilot shall automatically shut off the gas to all toaster burners including the toaster pilot burners, in the event means of ignition becomes inoperative (see 4.4.1).

3.4.5 Legs. When specified (see 6.2), type I toasters shall be provided with four removable legs. All type II toasters shall have four removable legs. The legs shall be of such length as to provide a minimum space of 2-5/8 inches below the base of the toaster.

3.4.6 Mounting. For shipboard use (see 6.2), all type I toasters shall have mounting bolt holes so that the toaster may be bolted down with 3/8-inch bolts. Spacing of the mounting bolt hole center lines shall be 11-1/2 inches (front to rear) by 17-5/16 inches (side to side).

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3.4.7 Lubrication. All bearings, revolving shafts, and other rotating or sliding parts requiring periodic lubrication shall be provided with means for lubrication and shall be lubricated prior to shipment. All grease fittings shall be accessible.

3.5 Operating requirements.

3.5.1 Type I toasters.

3.5.1.1 Voltages. Type I toasters shall be rated for one of the following nominal system voltages, as specified (see 6.2):

(a) 120-volt, direct current (DC)
(b) 120-volt, 1-phase, 60 hertz (Hz)
(c) 208-volt, 1-phase, 60 Hz
(d) 240-volt, 1-phase, 60 Hz
(e) 240-volt, DC

3.5.1.2 Electrical input. Maximum wattage input rating including both the motor and the heating elements shall not exceed 4.8 kilowatts (kw) for the size 2 and size 3 toasters.

3.5.2 Type II toasters.

3.5.2.1 Motors. Motors for type II toasters shall be designed to operate on a nominal 120-volt, 1-phase, 60 Hz, system.

3.5.2.2 Burners. The maximum total burner rating shall be 32,000 British thermal units (Btu) for the size 2 toasters.

3.6 Performance. Toasters shall be given the performance tests specified in 4.4.1 and 4.4.2. All parts, including thermostatic switches and motors, shall operate satisfactorily without failure and toasting shall continue during each cycle of this period. Toasters shall be capable of toasting 4-1/2 by 4-1/2-inch sliced bread, plus or minus 3/8 inch, and bun halves at a rate of 540 slices per hour for the size 2 toasters and 720 slices per hour for size 3 toaster. Slices of toast and bun halves produced during the performance test shall be evenly toasted and similar in appearance to each other when compared visually.

3.7 Marking. Identification and, if applicable, operating instructions shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or photosensitive. The plate thickness shall be not less than 0.012 inch.

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3.7.1 Identification. Each toaster identification plate shall include the manufacturer's model and serial number, name, trade name or trademark, of such known character as to be readily identifiable to the manufacturer. In addition, such information as required by UL 197 for type I toasters and ANSI Z21.31 and UL 197 for type II toasters. When required component identification number (toaster for Navy shipboard use only) shall also be included on the identification plate.

3.7.2 Users' instruction plate for type I toaster. The instruction plate for type I toasters shall be of sufficient overall size to contain, in legible characters, the following information:

TOASTER, ELECTRIC

USERS' INSTRUCTIONS

1. To start toaster, set "ON-OFF" switch to "ON" position.

2. Do not control heat by hand. Heat is controlled automatically.

3. For one side toasting, set control to "BUN" position. For two-side toasting, set control to "TOAST" position.

4. Set "ON-OFF" switch to "OFF" position when toaster is not in use.

5. Do not use a knife, fork, or other metal object to dislodge toast.

6. Disconnect electric plug before working on conveyor or other parts.

7. Use a small brush several times a day to remove crumbs clinging to wire basket and operating parts.

8. Keep toaster clean. For thorough cleaning of conveyor baskets, place in hot, soapy water and boil a few minutes, rinse and air dry.

9. Lubricate conveyor chains regularly with graphite suspended in penetrating oil in accordance with manufacturer's recommendations. For additional supply lubricant, requisition: Lubricating Oil, Collodial Graphite, 1-pint can, Federal Stock Number (NSN) 9150-00-227-0183.

NOTE: The above instructions shall be in characters not less than three-thirty-seconds of an inch high except the words "TOASTER, ELECTRIC" shall be in characters not less than three-sixteenths of an inch high. The words "USERS' INSTRUCTIONS" shall be in characters not less than one-eighth of an inch high.

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3.7.3 Users' instruction plate for type II toaster. The instruction plate for type II toasters shall be of sufficient overall size to contain, in legible characters, the following information:

TOASTER, GAS

USERS' INSTRUCTIONS

1. Apply heat and throw motor switch to start conveyor.

   Note: Do not shut off conveyor motor when burner is ignited.

2. Turn off heat when toaster is not in use. Let conveyor motor run for at least 10 minutes after burner flame is extinguished.

3. Do not control heat by hand. Heat is controlled automatically.

4. Do not use a knife, fork, or other metal objects to dislodge toast.

5. Disconnect electric plug before working on conveyor or other parts.

6. Use a small brush several times each day to remove crumbs clinging to the wire basket and operating parts.

7. Keep toaster clean. For thorough cleaning of conveyor baskets place in hot soapy water and boil a few minutes, rinse and air dry.

8. Lubricate conveyor chains regularly with graphite suspended in penetrating oil in accordance with manufacturer's recommendations. For additional supply lubricant, requisition: Lubricating Oil, Collodial Graphite, 1-pint can, Federal Stock Number (NSN) 9150-00-227-0183.

9. Discharge combustion products outdoors.

NOTE: The above instructions shall be in characters not less than three-thirty-seconds of an inch high except the words "TOASTER, GAS" shall be in characters not less than three-sixteenths of an inch high. The words "USERS' INSTRUCTIONS" shall be in characters not less than one-eighth of an inch high.

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3.8 Finish. The outer shell of all toasters shall be finished in accordance with the requirements of NSF No. 4. The surfaces of the finished toasters shall be free from rust, pits, stains, or discoloration.

3.9 Workmanship. All components and assemblies of the toasters shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings and molded parts if used, shall be free of sand, fins, pits, blowholes, and sprues. External surfaces shall be free from sharp edges and sharp corners.

3.9.1 Metal fabrication. Metal used in the fabrication of the toasters shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be free from trimming marks.

3.9.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free from cracks, burn holes, undercuts or incomplete fusion. All scale and flux shall be removed from the finish weld area.

3.9.3 Fastening devices. Holes punched or drilled shall be free of burrs. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets, when used, shall fill the hole completely and the heads shall be in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Process examination. When applicable, examination shall be made of the moisture and fungus resistance process to establish conformance with specified requirements (see (3.4.3.7). Whenever nonconformance is noted, correction shall be made to the process and all items processed.

4.2.3 End item inspection. The lot shall consist of all toasters of one type and size offered for inspection at one time. The sample unit shall be a completely fabricated toaster.

4.2.3.1 Visual examination. Examination shall be made of the toasters for defects classified in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Outer shell not finished in accordance with NSF Standard No. 4		X
	Evidence of discoloration rust, pits, or stains	X	
Construction and workmanship (general)	Component missing	X	
	Evidence of kinks		X
	Any burr, dent, slivers, sharp edges; rough die, tool, gouge, or grind marks	X	
	Any component fractured, buckled, dented, punctured, or malformed	X	
	Castings and molded parts when provided not free of sand, fins, pits, blowholes or sprues	X	
	Shell is not equipped with access panel on doors for repairs	X	
	Lubrication fitting(s) not accessible	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Conveyor rack	Functioning component inoperative	X	
	Not provided with manual override	X	
	Parts not lubricated (where required)	X	
	Parts insufficiently lubricated (where required)		X
Crumb tray	Not removable without tools	X	
Deflector	Not adjustable without tools	X	
Controls and connections	Not accessible to operator	X	
	Electrical connections not as specified	X	
	Not three-wire grounding type cord	X	
	Not three-prong grounding attachment plug	X	
Electric heating element	Not designed for only one method of installation	X	
	Like items not interchangeable	X	
Flexible metal gas line	Not type specified	X	
	Radius kinked	X	
	Hose not secured to back of toaster when disconnected from supply line	X	
Connector	Not provided	X	
Mounting holes (when applicable)	Missing	X	
Welding and soldering (when applicable)	Exposed surfaces not ground smooth	X	
	Corners not welded or soldered (as applicable)	X	
	Not free from oxide, scale, grease, or other foreign matter	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding and soldering (when applicable) (cont'd)	Not continuous, sound or smooth; or not free from flash, porosity, cracks, burn holes, incomplete fusion, undercut, or deformation of material	X	
	Excessive flux or solder not removed	X	
Threaded fasteners and rivets (as applicable)	Rivet not fully peened or upset	X	
	Rivet head not in full contact with riveted members	X	
	Missing, broken, stripped or fractured	X	
	Not drawn up tight	X	
Marking identification	Missing, incomplete or illegible	X	
Identification marking and users' instruction plate	Missing, incomplete, illegible, not proper location, not as specified		X

4.2.3.2 Dimensional examination. Inspection shall be made of the toasters to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be level S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

4.2.3.3 Testing of the end item. Each toaster shall be tested as specified in 4.4.1. Failure to pass the test shall be cause for rejection of the unit. One representative production unit shall be tested as specified in 4.4.2 and, when applicable, as specified in 4.4.5. Failure to pass either test shall be cause for rejection. Each sample toaster shall be tested as specified in 4.4.3 and 4.4.4 using an inspection level of S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

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4.2.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking complies with section 5 requirements. Defects shall be as set forth in table II. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with AQL of 2.5 defects expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing, damaged, or not as specified
Workmanship	Inadequate closure of case liners (when applicable) Containers bulged or distorted

4.2.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-591, PPP-B-601, or PPP-B-621, examination for defects in closure and strapping shall be in accordance with the applicable container specification.

4.2.4.2 Examination of interior packages. When interior packages are required to be in accordance with PPP-B-636, examination for defects in closure shall be in accordance with the appendix of PPP-B-636.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval, labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements, shall be examined and validated.

4.4 Tests. Any noncompliance with the specified requirements shall constitute failure of the test.

4.4.1 Operational. The toaster shall be operated for 30 minutes and during this period the heating and electrical mechanisms shall be checked for proper operation to determine compliance with 3.6. All type II toasters shall be checked for gas leaks and to determine compliance with 3.4.4.1.

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4.4.2 Functional test. The toaster shall be turned on and allowed to preheat for 10 minutes. Bread slices and bun halves shall then be continuously toasted for 30 minutes with the setting in a medium position. The toast and bun halves produced during this period shall be inspected to determine that satisfactory toast has been produced as specified in 3.4.3.4 and at the required rate as specified in 3.6.

4.4.3 Interference test. Position by hand the left conveyor drive sprockets inboard as far as possible. Position by hand the bread racks as far as possible to the left. Rotate the conveyor by hand for one complete revolution to determine compliance with 3.4.2. Any interference between the bread rack wires and the conveyor studs shall constitute failure of the test. Repeat above test for the right conveyor chain by positioning the right conveyor chain sprockets inboard as far as possible and the bread racks as far as possible to the right.

4.4.4 Overrunning clutch test. With the toaster inoperative, the conveyor rack assembly shall be rotated one complete revolution manually to determine compliance with 3.4.2.

4.4.5 Electromagnetic compatibility test. When electromagnetic compatibility is required (see 3.4.3.6) the toaster shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing organization. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, and the Electromagnetic Test Report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Prepackaging. Removable legs, protruding parts, and other components that might be damaged in shipment shall be disassembled from the toaster. Component parts of each toaster that are free to move, such as the removable pans, shall be secured in a manner to prevent movement while in transit.

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5.1.1.2 Packaging. Each toaster, together with disassembled components, shall be packaged in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 275, of PPP-B-636. The contents shall be securely held in position within the box by folded, built-up or die-cut pads made of the same material as the box or by wood braces, or by a combination of those materials in a manner to prevent movement while in transit. Each box shall be closed in accordance with method II in the appendix of PPP-B-636.

5.1.1.3 Level B. Toasters shall be prepackaged as specified in 5.1.1.

5.1.2 Level C. The toasters shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each toaster, packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Each shipping container shall be provided with type I or II, grade C, sealed case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix of the applicable container specification. Alternatively the toasters may be packed in accordance with the level A requirements of PPP-P-600.

5.2.2 Level B. Toasters, packaged as specified in 5.1.1 and not exceeding 90 pounds gross weight, shall not require any further packing. When the gross weight exceeds 90 pounds, each packaged toaster shall be packed in a snug-fitting shipping container conforming to class I, style A or B of PPP-B-591; domestic type, style A or B of PPP-B-601; or class 1, style 2 or 4 of PPP-B-621. Alternatively, the toasters may be packed in accordance with the level B packing requirements of PPP-P-600.

5.2.3 Level C. Toasters, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

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5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The toasters covered by this specification are intended for use where volume food serving is done. Type II toasters are not intended for shipboard use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2).
(c) When a nonadjustable deflector for loading side discharge is required (see 3.4).
(d) If the terminal board or box is not required on the type I toasters (see 3.4.3.2).
(e) When a switch is not required for the rear heating element (see 3.4.3.4).
(f) When electromagnetic compatibility is required (see 3.4.3.6).
(g) When fungus and moisture treatment of electrical components is required (see 3.4.3.7).
(h) When a quick-disconnect assembly is required for type II toasters (see 3.4.4).
(i) When other than 60 inch length of gas-line connector is required (see 3.4.4).
(j) When legs are required for type I toasters (see 3.4.5).
(k) Whether toasters are intended for shipboard use (see 3.4.6).
(l) Voltage on type I toasters (see 3.5.1.1).
(m) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(n) Marking required (see 5.3).

6.3 Contract data requirements. Any data items to be delivered under any contract for the items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

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6.4 Supersession data. Toasters identified as "size 1" in Federal Specification S-T-540D, dated April 17, 1972 have been eliminated in this revision due to no suppliers.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - MD, ME
Navy - MC, SA

User activity:

Army - CE

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS
VA - DMS

Project No. 7310-0527

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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